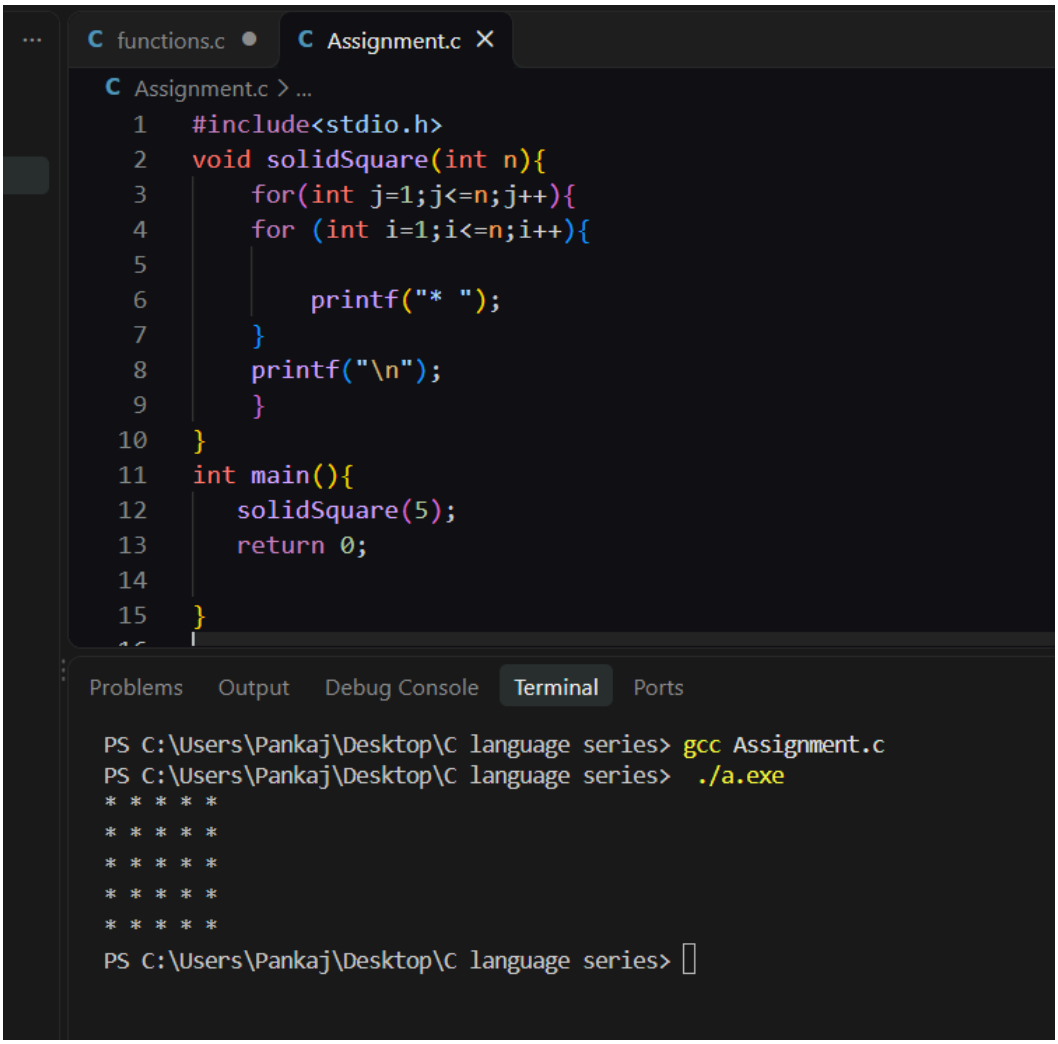
Module 7 — Nested-loop and pattern-printing questions

Question 1 — Print a Solid Square

Problem statement: Given a positive integer n, print a square containing n rows and n columns of asterisks (*). Do not print spaces between asterisks.

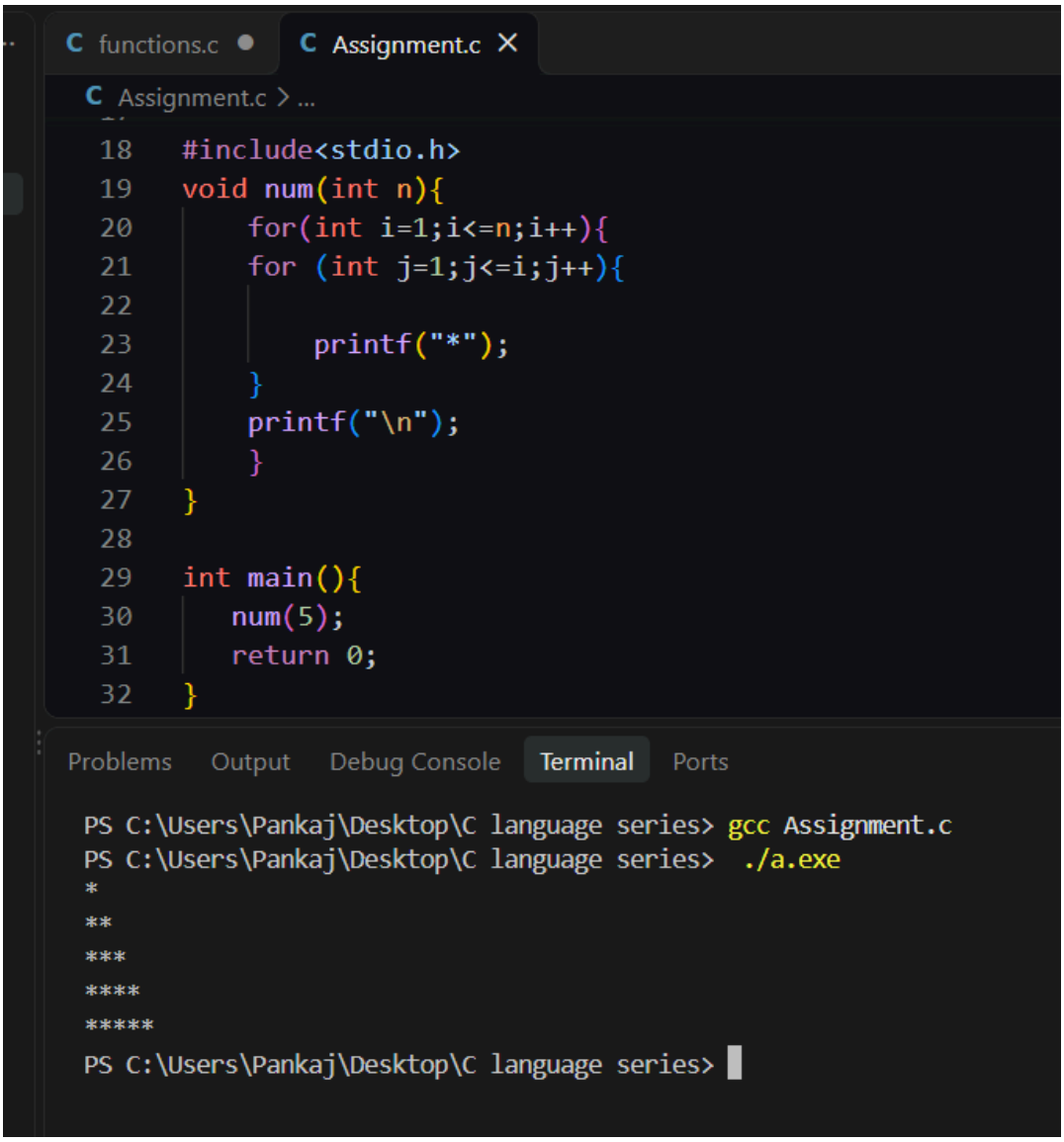
[Source Code]



Question 2 — Print a Right-Angled Triangle

Problem statement: Given n, print a left-aligned triangle. Row i must contain exactly i asterisks, for i from 1 to n.

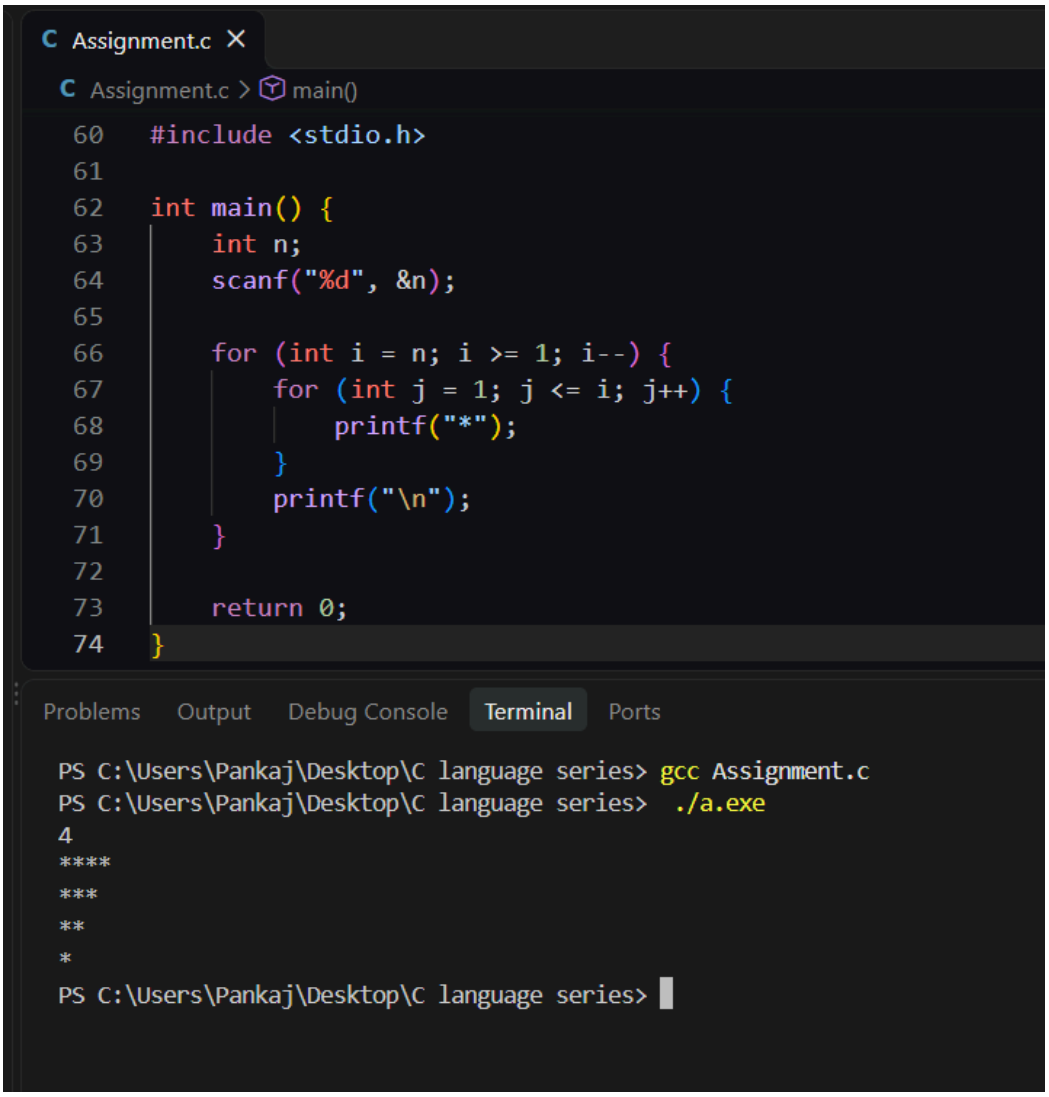
[Source Code]



Question 3 — Print an Inverted Triangle

Problem statement: Given n, print a left-aligned inverted triangle. The first row contains n asterisks; each following row contains one fewer.

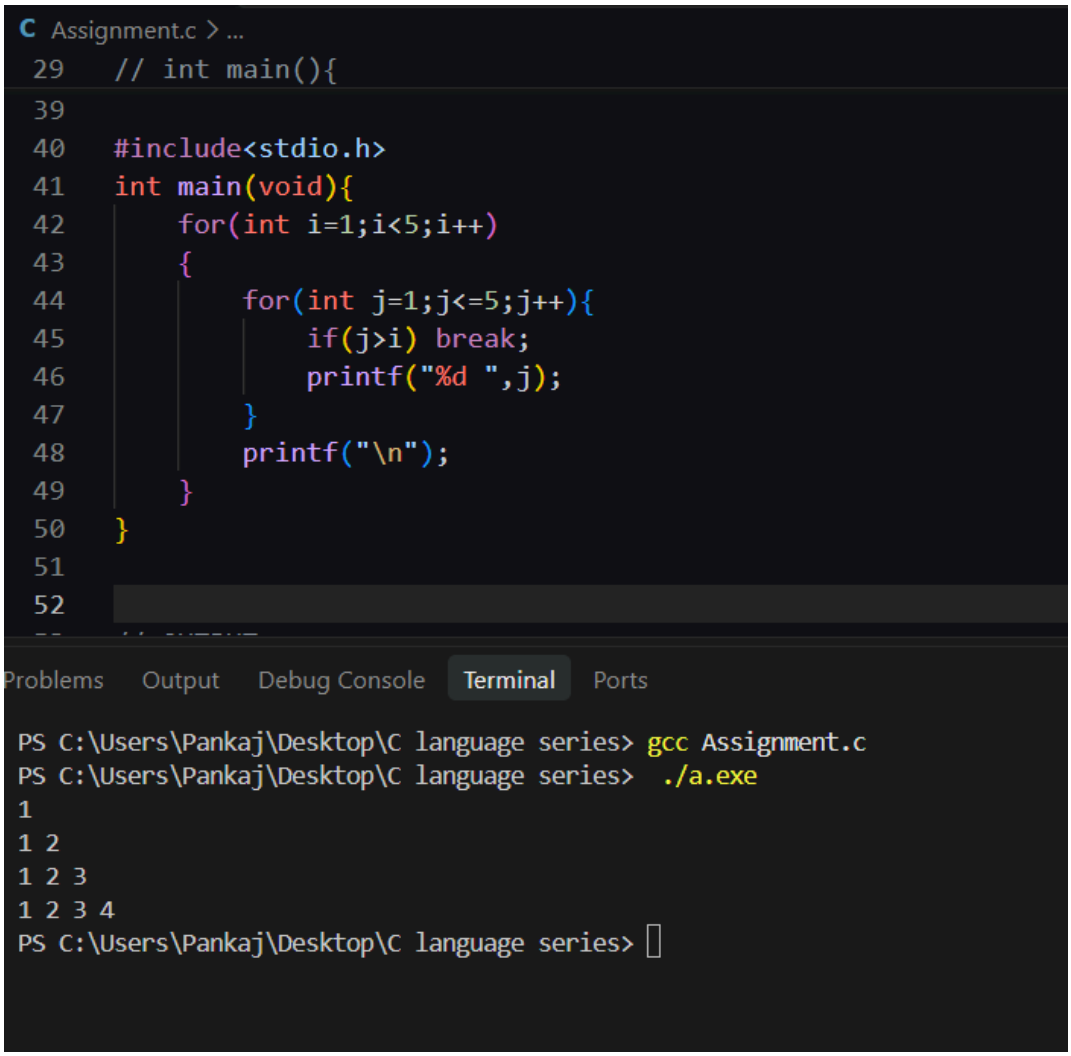
[Source Code]



Question 4 — Print a Number Triangle

Problem statement: Given n, print n rows. On row i, print integers from 1 through i, separated by one space.

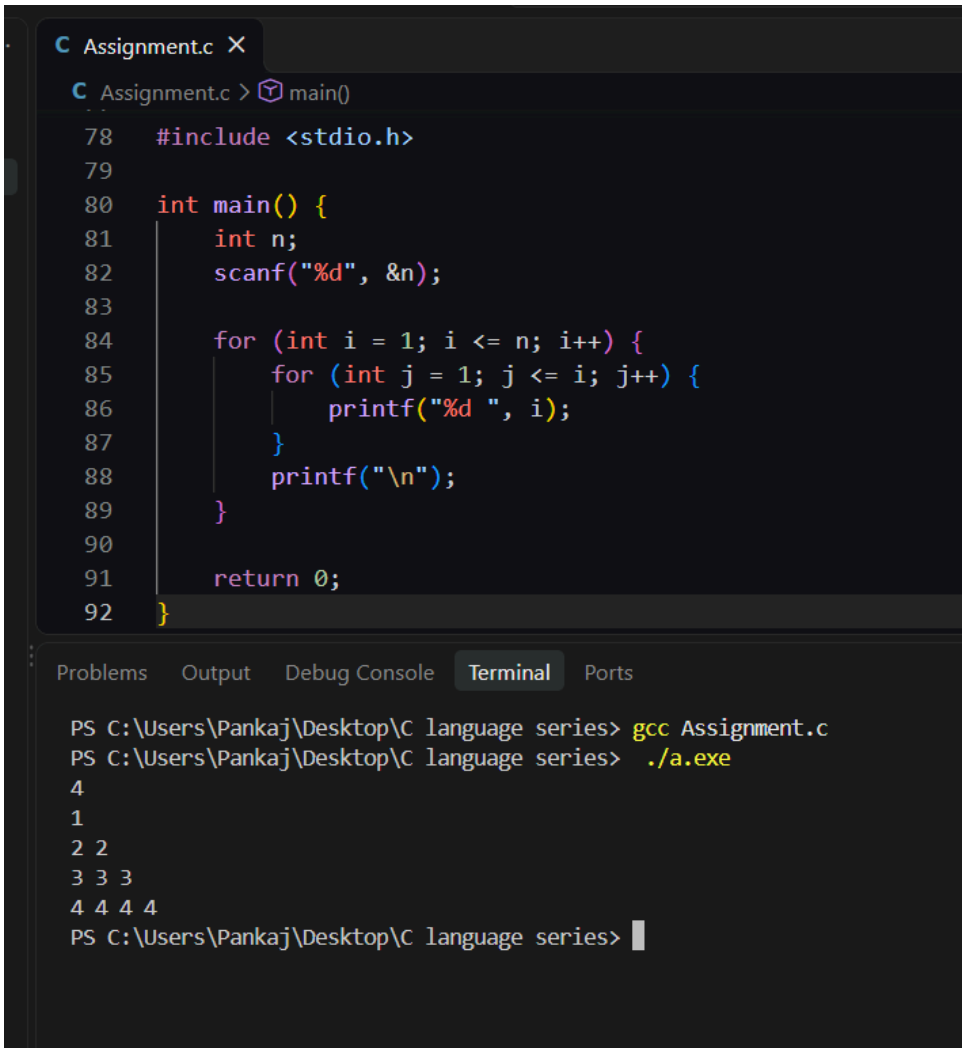
[Source Code]



Question 5 — Print a Repeated-Number Triangle

Problem statement: Given n, print n rows. On row i, print the number i exactly i times, separated by one space.

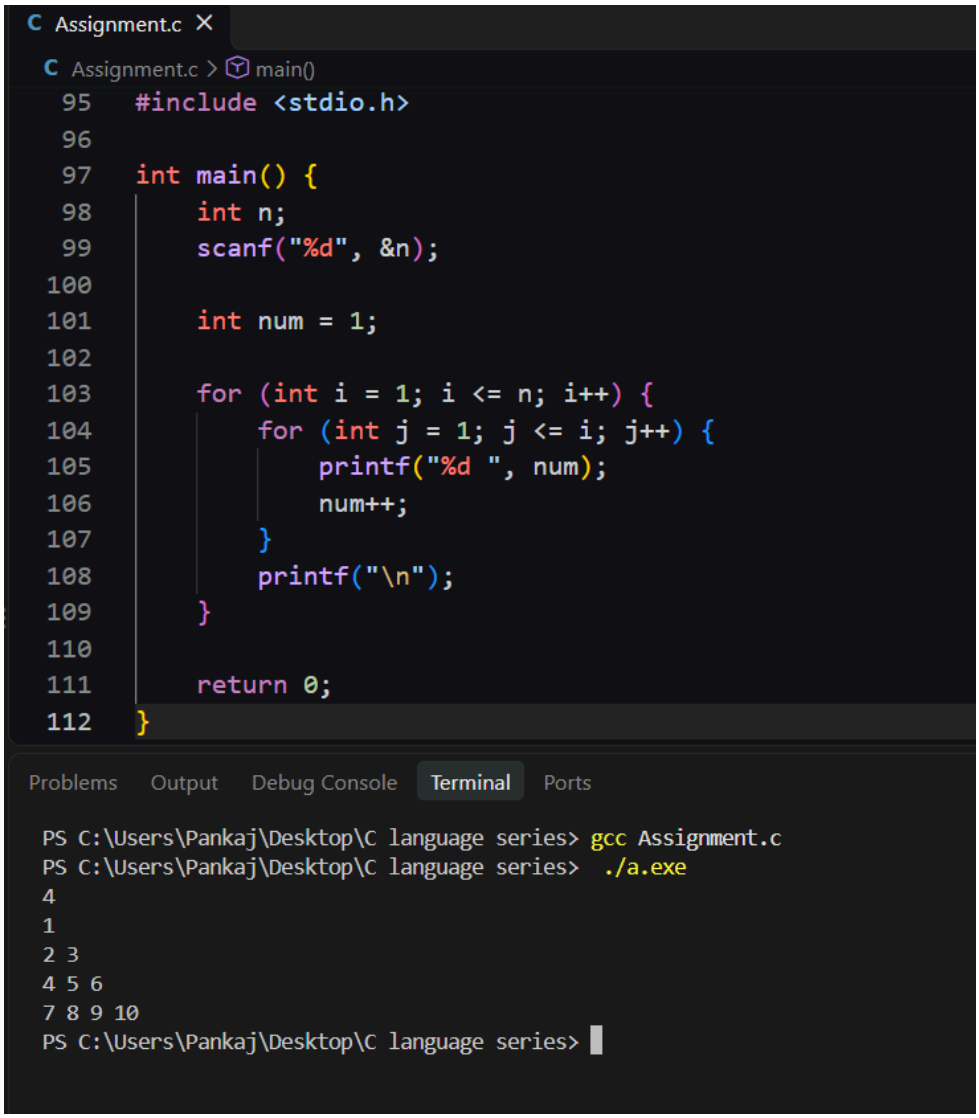
[Source Code]



Question 6 — Print a Consecutive-Number Triangle

Problem statement: Print consecutive positive integers across triangular rows. Row 1 has one number, row 2 has two, and so on. Counting starts at 1 and continues across rows.

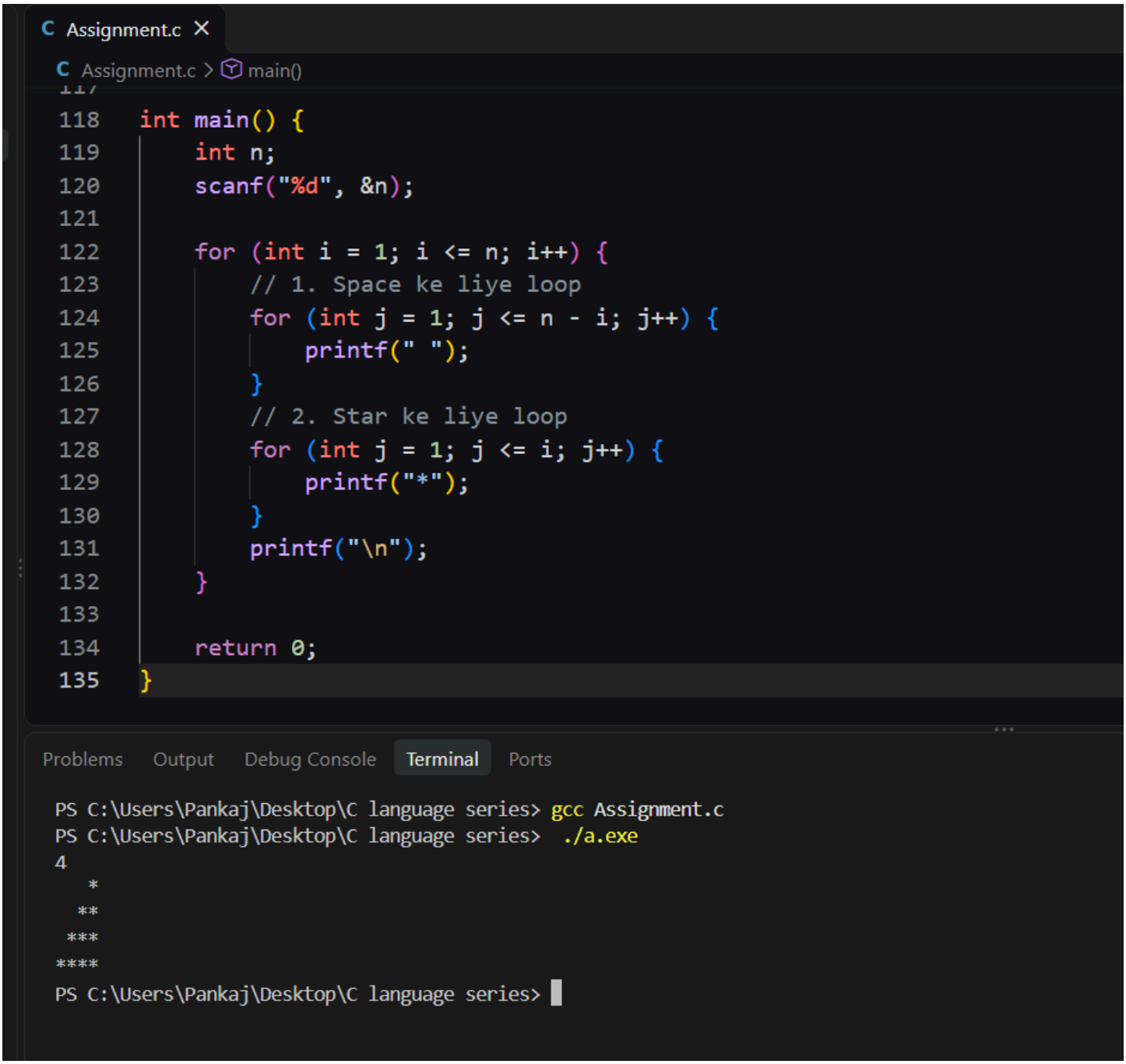
[Source Code]



Question 7 — Print a Right-Aligned Star Triangle

Problem statement: Given n, print a right-aligned triangle. Row 1 has one asterisk and row n has n asterisks. Use leading spaces to align the right edges.

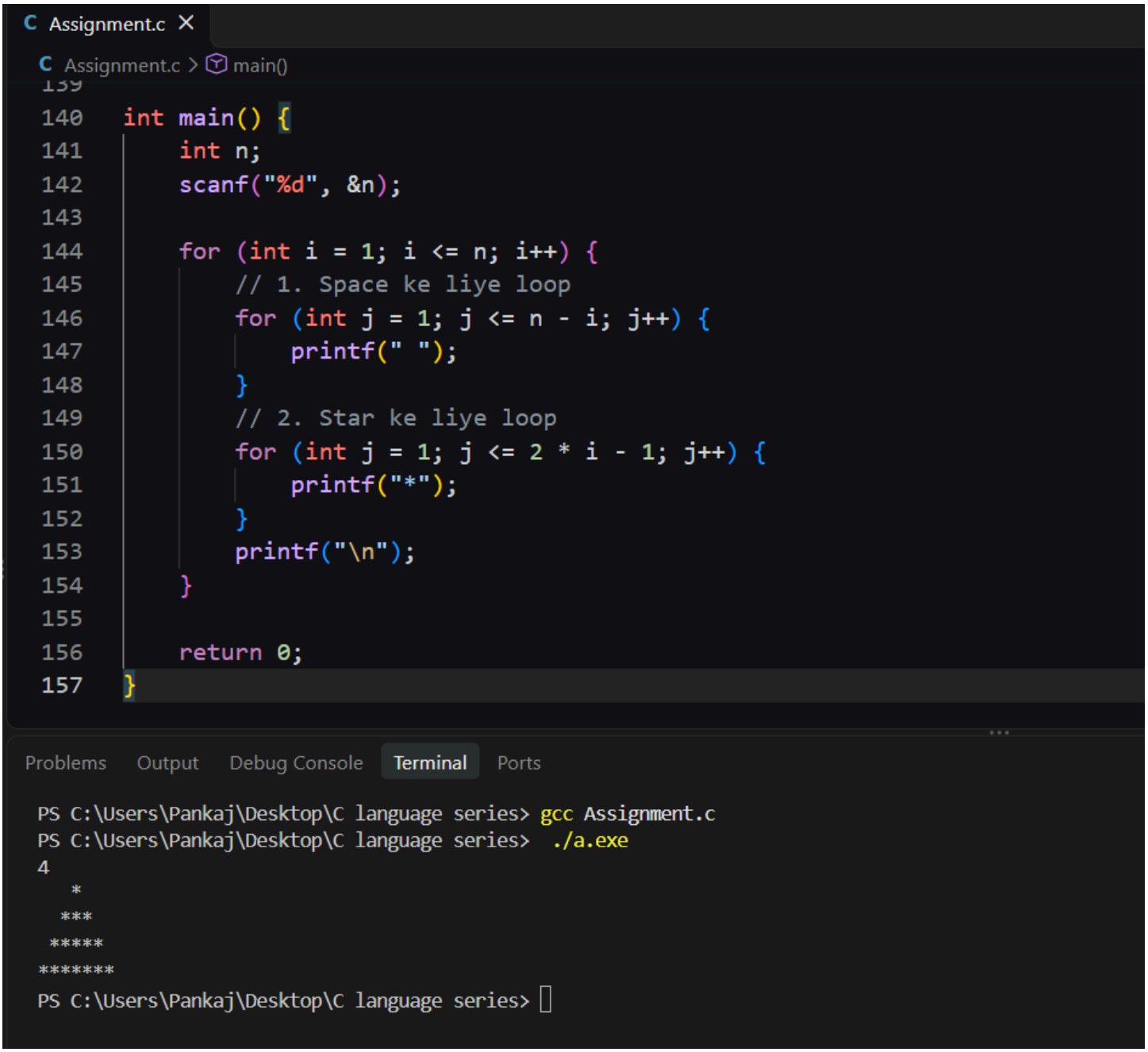
[Source Code]



Question 8 — Print a Centered Pyramid

Problem statement: Given n, print a centered pyramid with n rows. Row i contains 2i - 1 asterisks, preceded by enough spaces to center it.

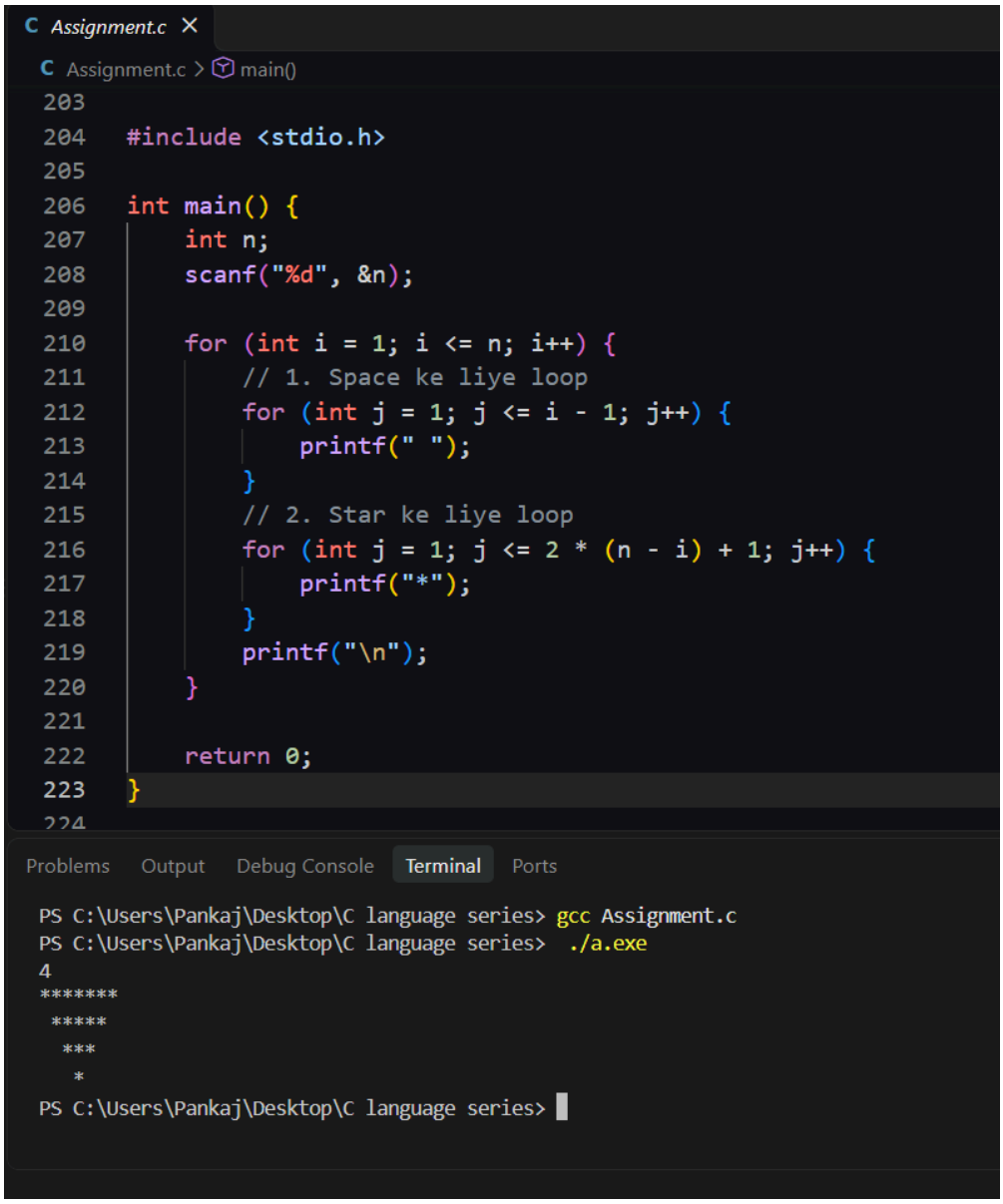
[Source Code]



Question 9 — Print an Inverted Centered Pyramid

Problem statement: Given n, print an inverted centered pyramid. The first row contains 2n - 1 asterisks; each next row has two fewer and one more leading space.

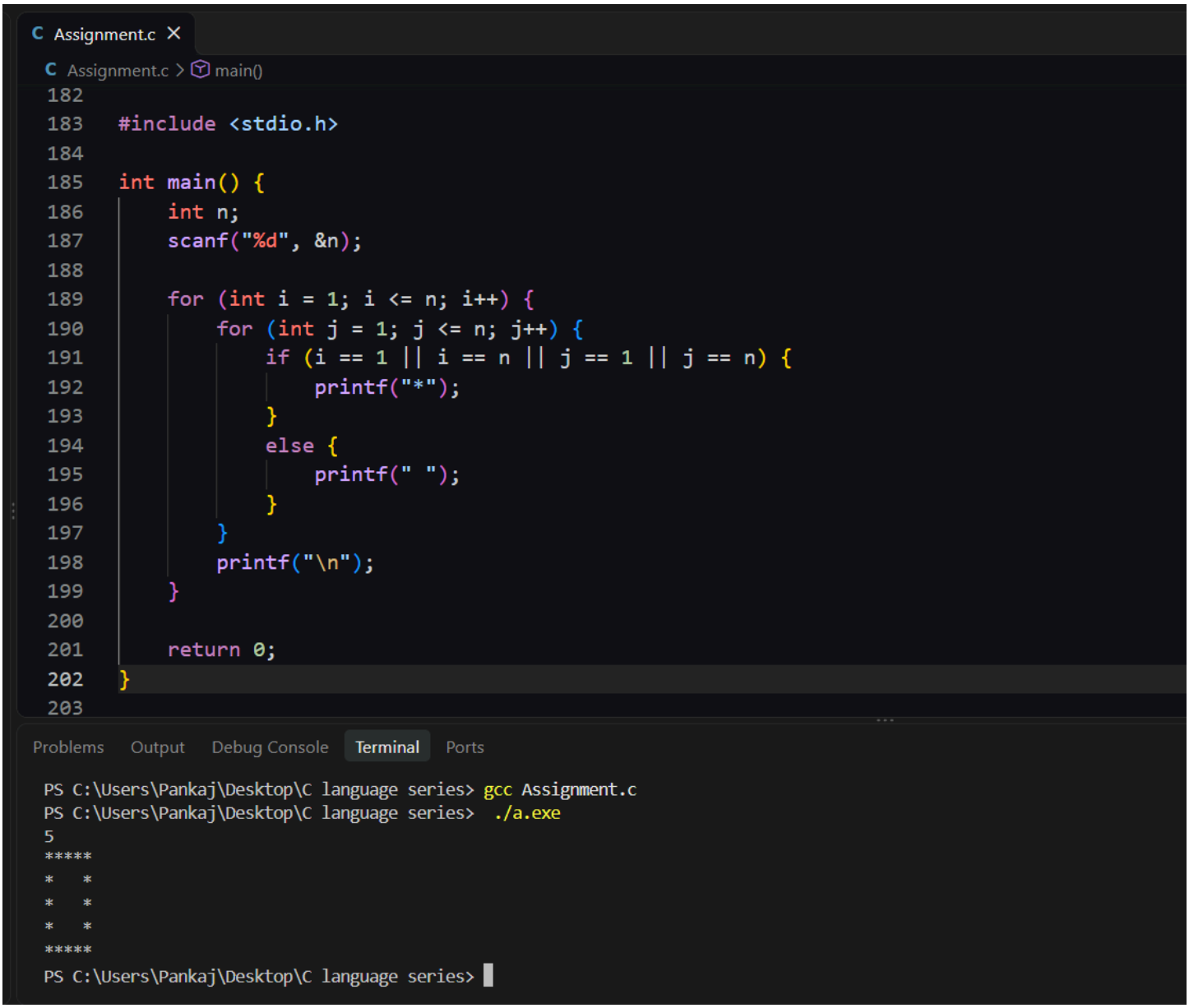
[Source Code]



Question 10 — Print a Hollow Square

Problem statement: Given n, print an n-by-n hollow square. Print an asterisk on the first row, last row, first column, or last column. Print a space inside.

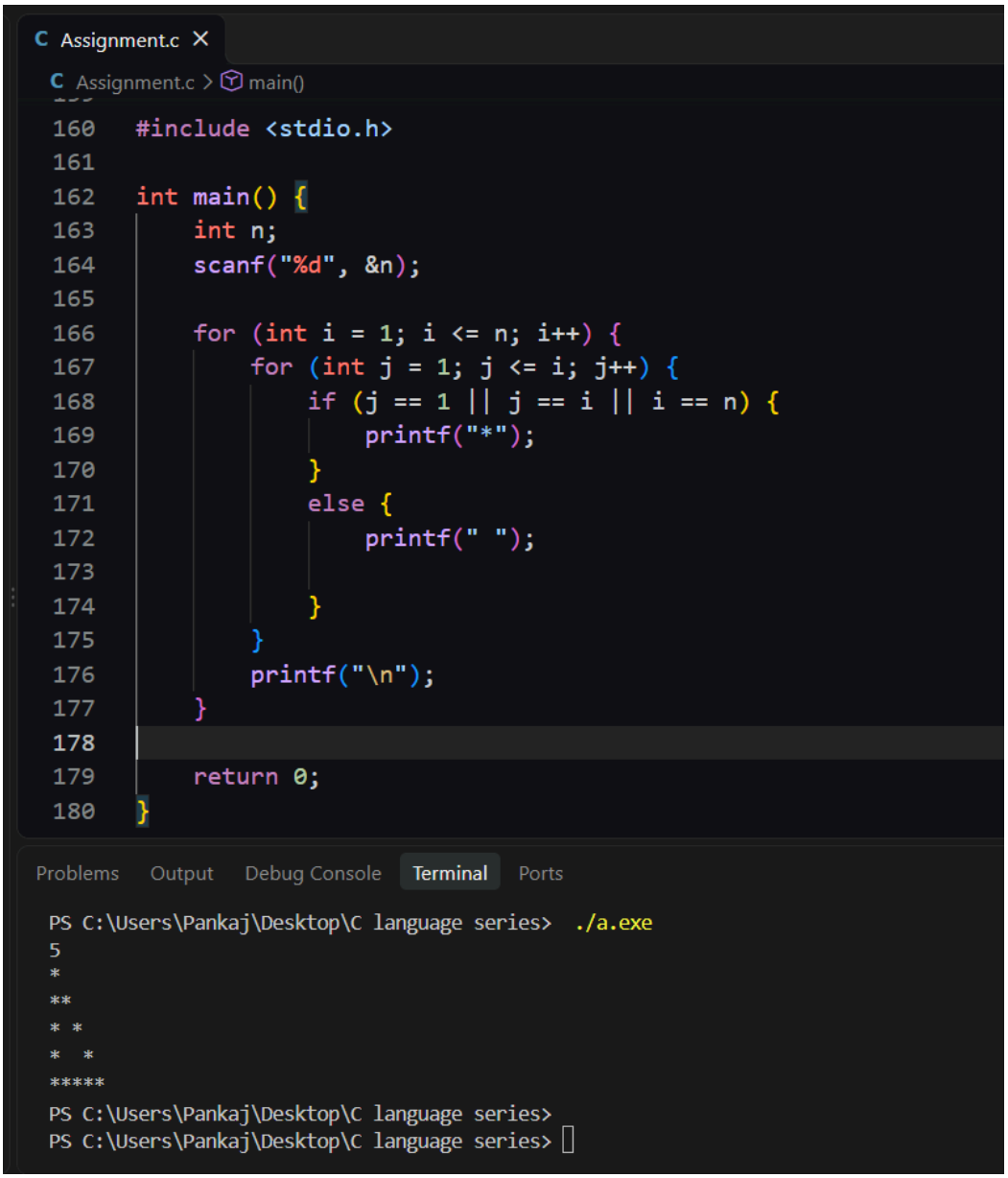
[Source Code]



Question 11 — Print a Hollow Right-Angled Triangle

Problem statement: Given n, print the outline of a left-aligned right-angled triangle. Print asterisks in the first column, on the diagonal boundary, and across the final row. Print spaces inside.

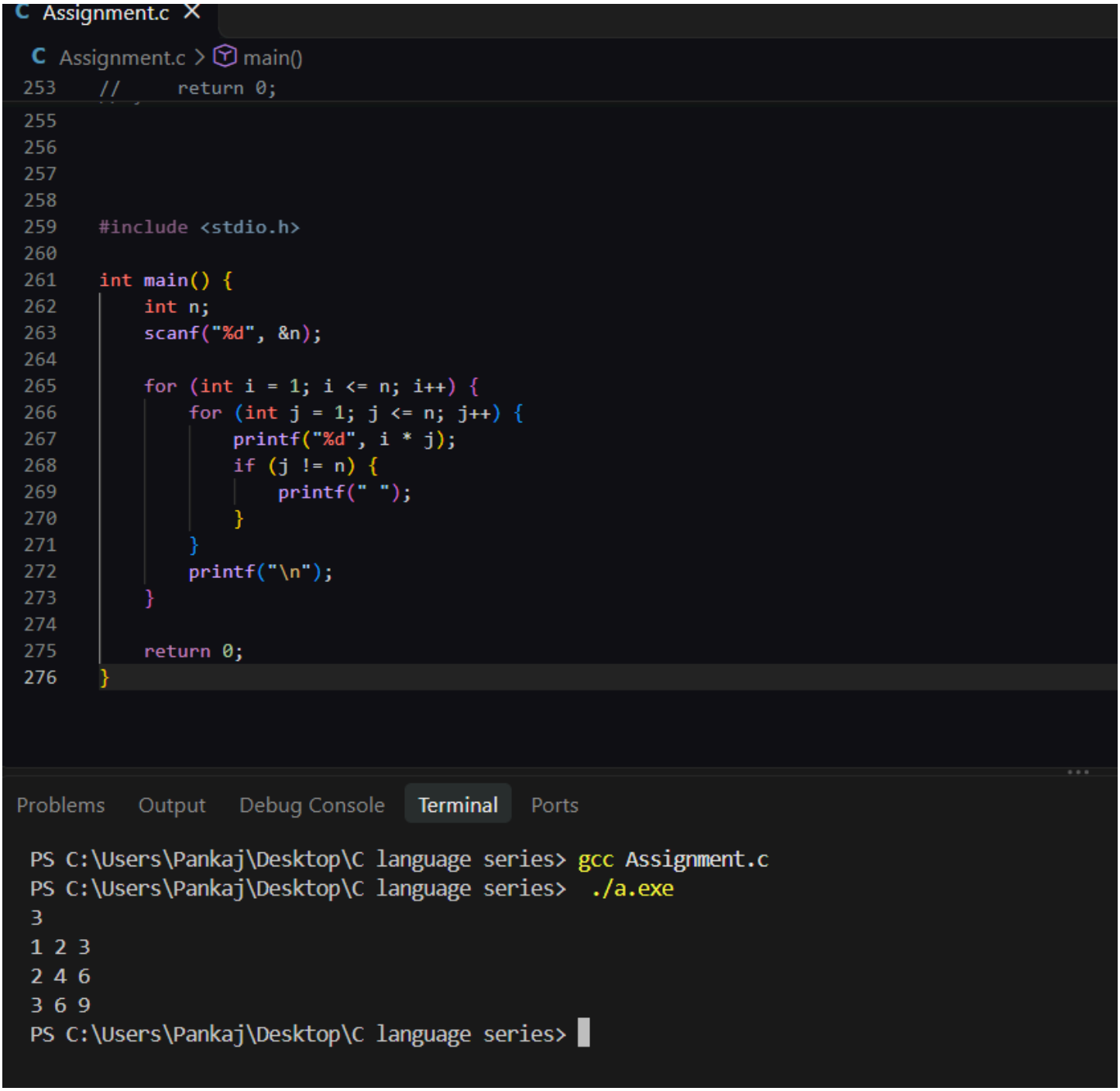
[Source Code]



Question 12 — Print a Multiplication Table Grid

Problem statement: Given n, print an n-by-n multiplication grid. At row i and column j, print i multiplied by j. Separate values with one space.

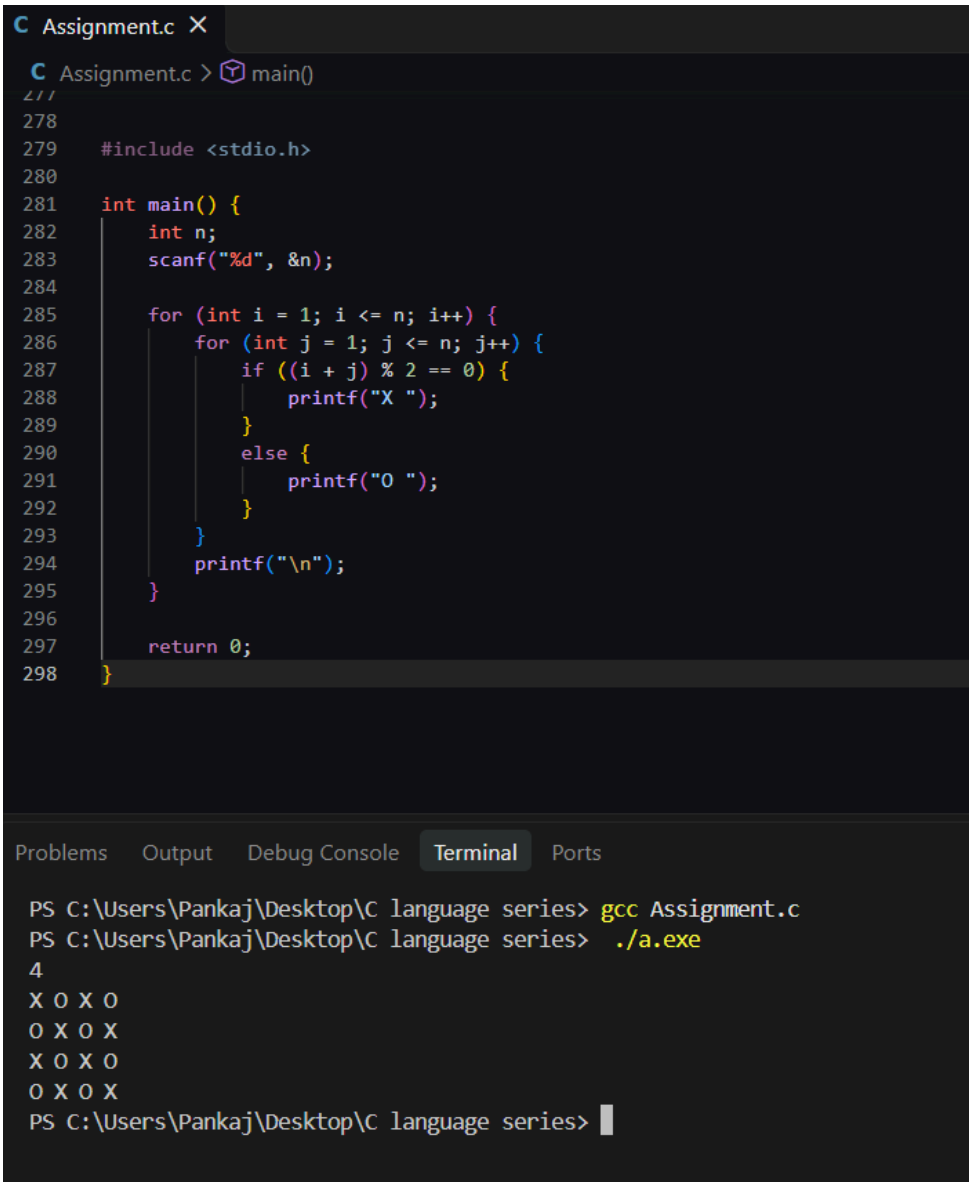
[Source Code]



Question 13 — Print a Checkerboard

Problem statement: Given n, print an n-by-n checkerboard using X and O. The top-left character must be X; adjacent horizontal and vertical cells alternate.

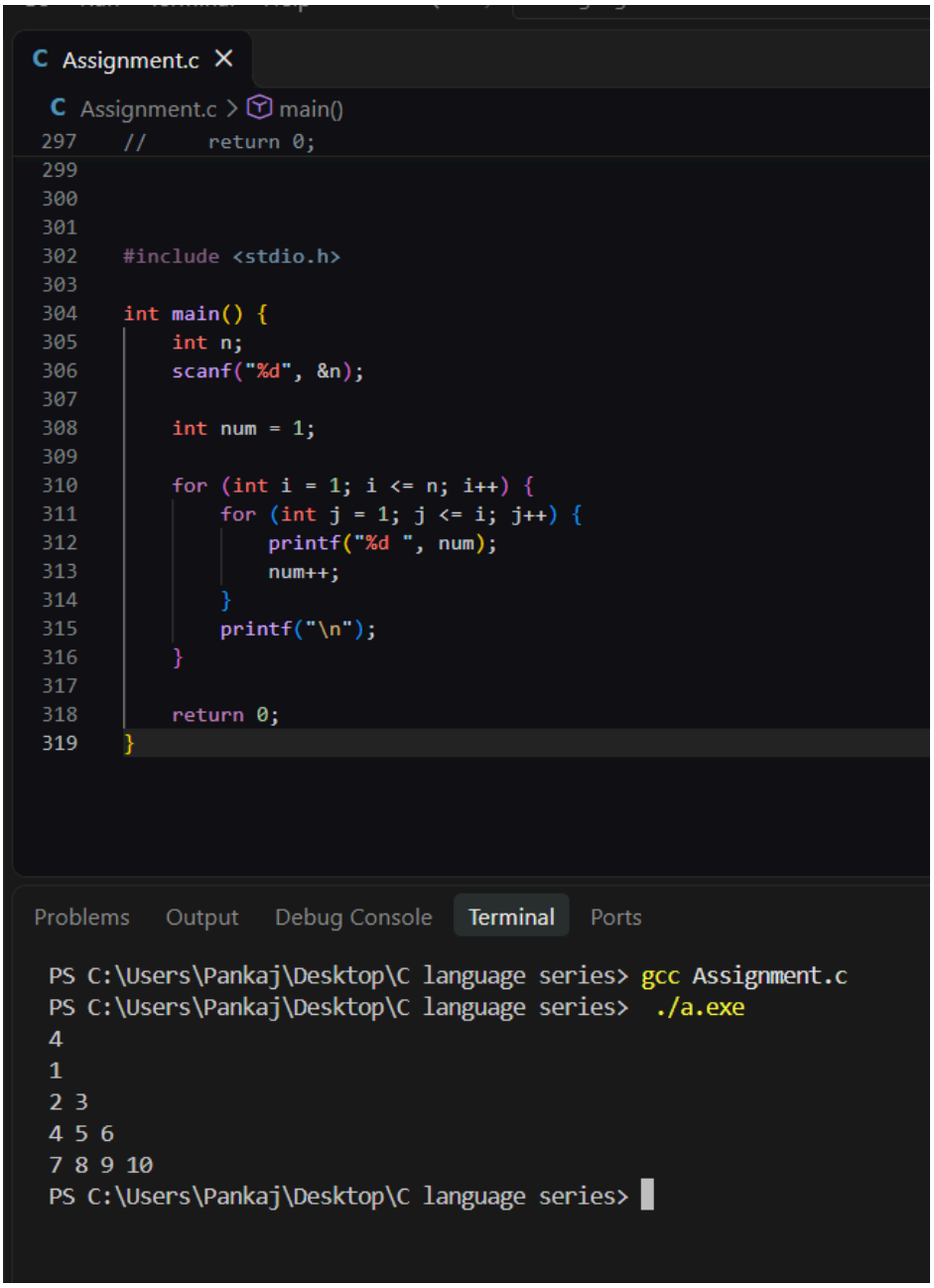
[Source Code]



Question 14 — Print Floyd's Triangle

Problem statement: Given n, print Floyd's triangle with n rows. Fill it with consecutive positive integers starting at 1, placing one number on row 1, two on row 2, and so on.

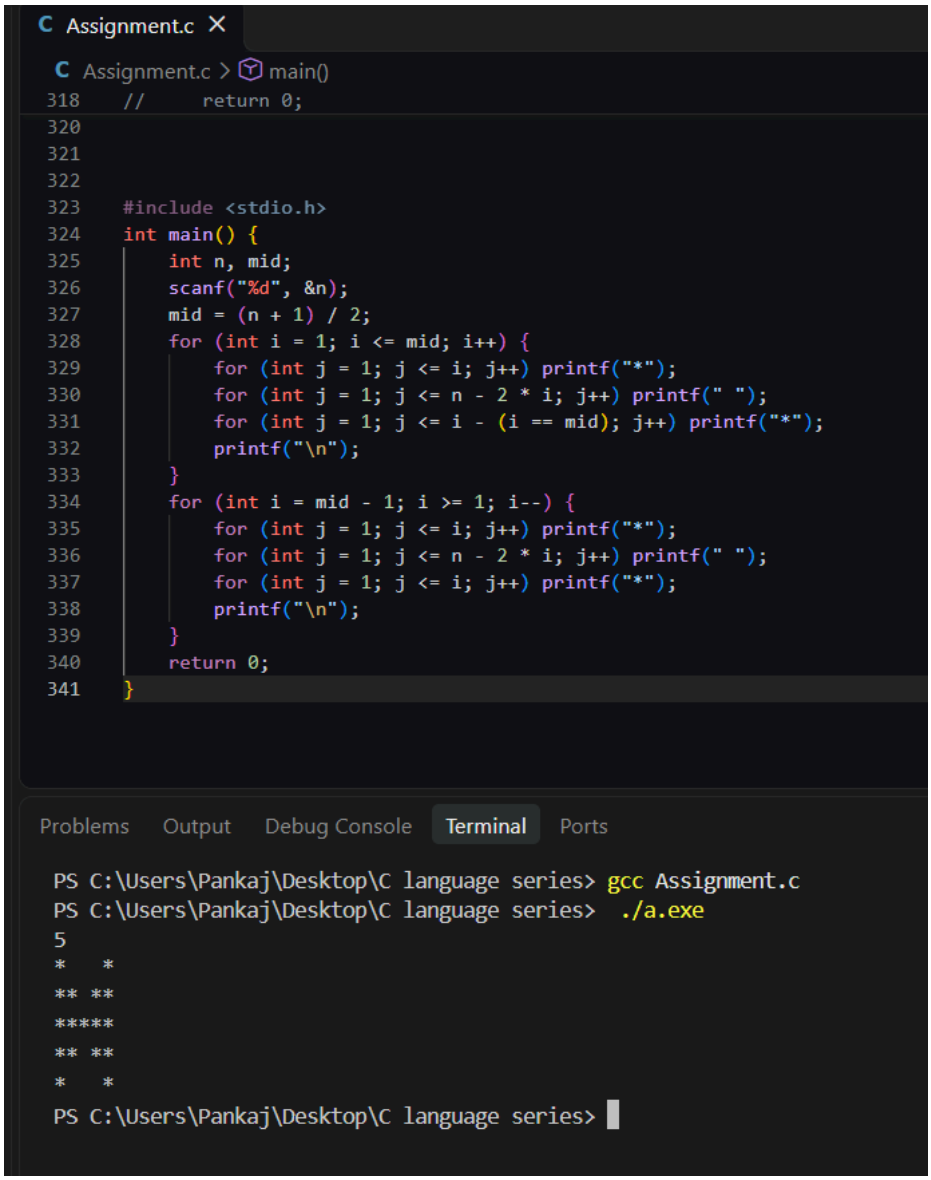
[Source Code]



Question 15 — Print a Butterfly Pattern

Problem statement: Given an odd integer n, print a butterfly made of asterisks. The top half widens row by row and the lower half mirrors it. Use spaces between the left and right wings.

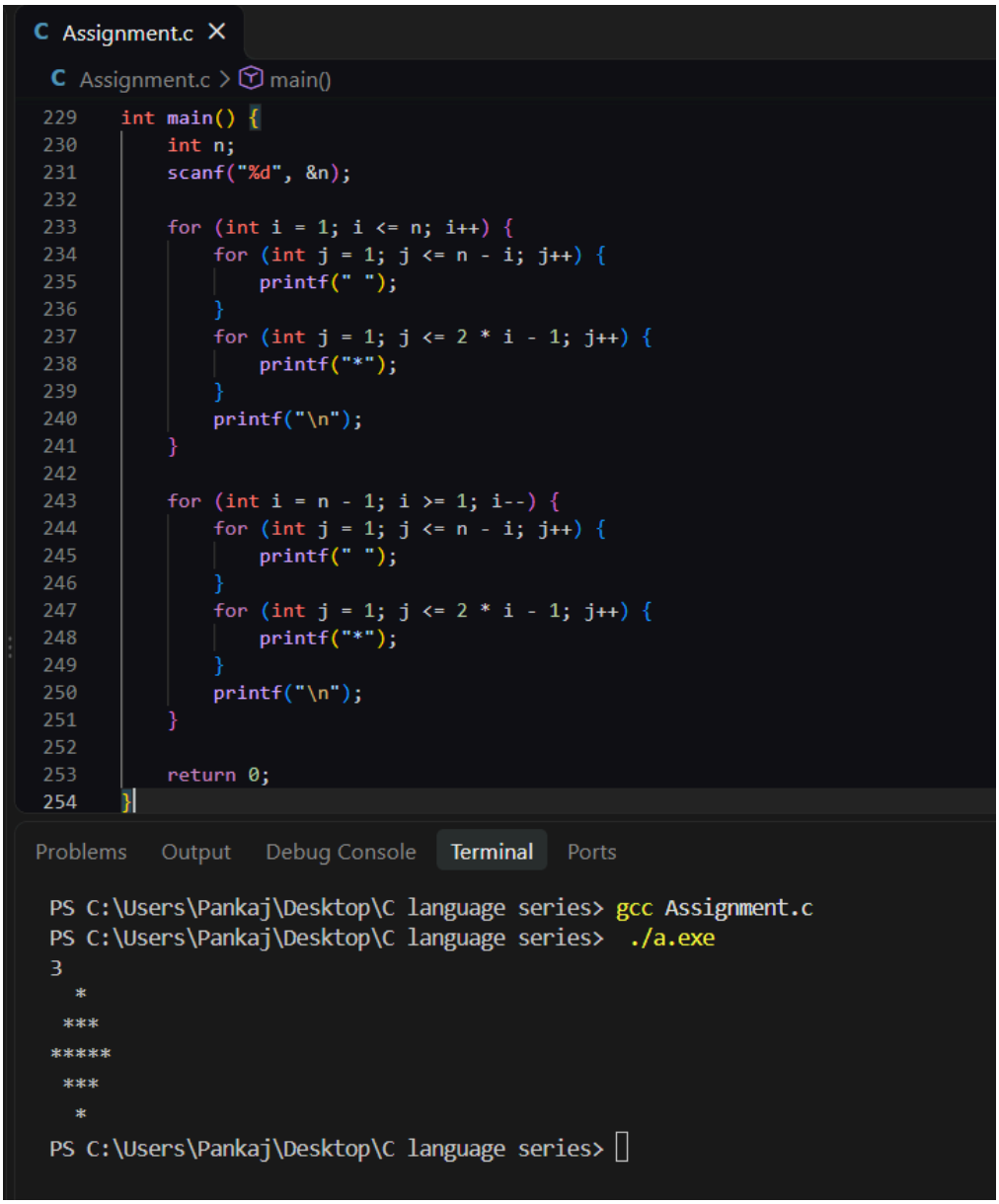
[Source Code]



Question 16 — Print a Diamond Pattern

Problem statement: Given n, print a centered diamond. The upper half has n rows including the widest row; then print n - 1 rows in decreasing width.

[Source Code]


Assignment.c
Assignment.c > main()
int main() {
int n;
scanf("%d", &n);
for (int i = 1; i <= n; i++) {
for (int j = 1; j <= n - i; j++) {
printf(" ");
}
for (int j = 1; j <= 2 * i - 1; j++) {
printf("*");
}
printf("\n");
}
for (int i = n - 1; i >= 1; i--) {
for (int j = 1; j <= n - i; j++) {
printf(" ");
}
for (int j = 1; j <= 2 * i - 1; j++) {
printf("*");
}
printf("\n");
}
return 0;
}
Problems
Output
Debug Console
Terminal
Ports
PS C:\Users\Pankaj\Desktop\C language series> gcc Assignment.c
PS C:\Users\Pankaj\Desktop\C language series> ./a.exe
3
*
***
*****
***
*
PS C:\Users\Pankaj\Desktop\C language series>